Deep learning

2.1. Loss and risk

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The general objective of machine learning is to capture regularity in data to make predictions.

In our regression example, we modeled age and blood pressure as being linearly related, to predict the latter from the former.

There are multiple types of inference that we can roughly split into three categories:

- Classification (e.g. object recognition, cancer detection, speech processing),
- regression (e.g. customer satisfaction, stock prediction, epidemiology), and
- density estimation (e.g. outlier detection, data visualization, sampling/synthesis).

Notes

The content of this lesson is not specific to deep learning but is fundamental: overfitting, capacity, embeddings, etc.
We can classify inference methods in roughly three categories:

- In classification, we get a signal and we want to predict a discrete label, e.g.

Task	Input	Output
Object recognition	Image	Class label ("cat", "dog", etc.)
Cancer detection	Gene expression	"cancer" or "no cancer"
Speech processing	Sound sample	Word or phonem

- In regression, we get as input a signal and we want to predict a continuous quantity. Contrary to classification there is a notion of metric on the value space.

Task	Input	Output
Customer Satisfaction	Questionnaire	Satisfaction $\in [0, 5]$
Stock prediction	Past exchange rate	Exchange rate tomorrow

- In density estimation, we try to capture the structure of the data, instead of predicting a single value.

Task	Input	Output
Outlier detection	Data point	Is unlikely under the model
Image generation	Gaussian noise image	Synthetic realistic image

The standard formalization for classification and regression considers a measure of probability

$$\mu_{X,Y}$$

over the observation/value of interest, and i.i.d. training samples

$$(x_n, y_n), \ n = 1, \ldots, N,$$

and for density estimation

$$\mu_X$$

and

$$x_n, \ n = 1, \ldots N.$$

Intuitively, for classification a often intuitive interpretation is

$$\mu_{X,Y}(x, y) = \mu_{X|Y=y}(x)\, P(Y = y)$$

that is, draw Y first, and given its value, generate X.

So the conditional distribution

$$\mu_{X|Y=y}$$

stands for the distribution of the observable signal for the class y (e.g. "sound of an /ē/", "image of a cat").

For regression, one would interpret the joint law more naturally as

$$\mu_{X,Y}(x,y) = \mu_{Y|X=x}(y)\,\mu_X(x)$$

which would be: first, generate X, and given its value, generate Y.

In the simple cases

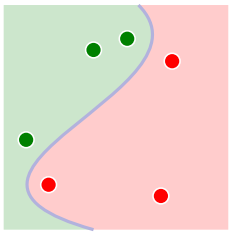
$$Y = f(X) + \epsilon$$

where f is the deterministic dependency between x and y (e.g. affine), and ϵ is a random noise, independent of X (e.g. Gaussian).

With such a probabilistic perspective, we can more precisely define the three types of inferences we introduced before:

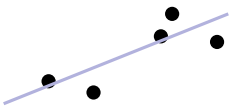
Classification,

- (X, Y) random variables on $\mathcal{Z} = \mathbb{R}^D \times \{1, \ldots, C\}$,
- we want to estimate $\text{argmax}_y P(Y = y \mid X = x)$.



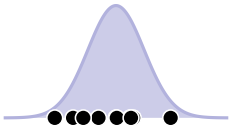
Regression,

- (X, Y) random variables on $\mathcal{Z} = \mathbb{R}^D \times \mathbb{R}$,
- we want to estimate $\mathbb{E}(Y \mid X = x)$.



Density estimation,

- X random variable on $\mathcal{Z} = \mathbb{R}^D$,
- we want to estimate μ_X.



Notes

The formulations above are for the vanilla and simplest cases of classification, regression, and density estimation.
In classification, what we want is to find, given a sample $x \in \mathbb{R}^D$, the most likely class y that the sample belongs to.
For instance, in the case of image classification, many models process an input image $D \sim 3 \times 224 \times 224$, and there can be $C = 1000$ classes.
In regression, the quantity to predict can be in high dimension and the target value not be the conditional expectation but for instance a more robust value that ignore long tails in the distribution.

The boundaries between these categories are fuzzy:

- Regression allows to do classification through class scores.
- Density models allow to do classification thanks to Bayes' law.

etc.

Risk, empirical risk

Learning consists of finding in a set $\mathscr{F}$ of functionals a "good" f^* (or its parameters' values) usually defined through a loss

$$\ell : \mathscr{F} \times \mathscr{Z} \to \mathbb{R}$$

such that $\ell(f, z)$ increases with how wrong f is on z. For instance

- for classification:
$$\ell(f, (x, y)) = \mathbf{1}_{\{f(x) \neq y\}},$$
- for regression:
$$\ell(f, (x, y)) = (f(x) - y)^2,$$
- for density estimation:
$$\ell(q, z) = -\log q(z).$$

The loss may include additional terms related to f itself.

Notes

$\mathscr{F}$ is the set of all the functions that the learning phase may produce. For instance, when the architecture of a neural network is fixed, $\mathscr{F}$ contains one mapping per parameter configuration. Note that this space is a continuous space if the parameter space is.
Learning consists of finding a "good" function, that is, a function that "does what is it supposed to do" (classifying, regressing, etc.)
The loss ℓ is a function that indicates how bad a functional f (e.g. classifier, regressor) is performing at its task on a sample z: The larger the loss, the worse the prediction of f on z.
As we will see later, in addition to terms related to the response of f, the loss function may contain elements relative to the structure of f, such as regularization terms, to control the magnitude of the parameters, the curvature, etc.

We are looking for an f with a small **expected risk**

$$R(f) = \mathbb{E}_Z\left(\ell(f, Z)\right),$$

which means that our learning procedure would ideally choose

$$f^* = \operatorname*{argmin}_{f \in \mathcal{F}} R(f).$$

Although this quantity is unknown, if we have i.i.d. training samples

$$\mathcal{D} = \{Z_1, \ldots, Z_N\},$$

we can compute an estimate, the **empirical risk**:

$$\hat{R}(f; \mathcal{D}) = \hat{\mathbb{E}}_{\mathcal{D}}(\ell(f, Z)) = \frac{1}{N}\sum_{n=1}^{N} \ell(f, Z_n).$$

Notes

The expected risk is the expectation of the loss when the data follows the true distribution of Z, and $R(f)$ is unknown because we do not have access to the true distribution of Z.

We have

$$\begin{aligned}
\mathbb{E}_{Z_1,\ldots,Z_N}\left(\hat{R}(f;\mathscr{D})\right) &= \mathbb{E}_{Z_1,\ldots,Z_N}\left(\frac{1}{N}\sum_{n=1}^{N}\ell(f,Z_n)\right) \\
&= \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}_{Z_n}\left(\ell(f,Z_n)\right) \\
&= \mathbb{E}_{Z}\left(\ell(f,Z)\right) \\
&= R(f).
\end{aligned}$$

The empirical risk is an **unbiased estimator** of the expected risk.

Finally, given $\mathscr{D}$, $\mathscr{F}$, and ℓ, "learning" aims at computing

$$f^* = \operatorname*{argmin}_{f \in \mathscr{F}} \hat{R}(f; \mathscr{D}).$$

- Can we bound $R(f)$ with $\hat{R}(f; \mathscr{D})$?

  Yes if f is not chosen using $\mathscr{D}$. Since the Z_n are independent, we just need to take into account the variance of $\hat{R}(f; \mathscr{D})$.

- Can we bound $R(f^*)$ with $\hat{R}(f^*; \mathscr{D})$?

  Unfortunately not simply, and not without additional constraints on $\mathscr{F}$.

  For instance if $|\mathscr{F}| = 1$, we can!

Note that in practice, we call "loss" both the functional

$$\ell : \mathscr{F} \times \mathscr{Z} \to \mathbb{R}$$

and the empirical risk minimized during training

$$\mathscr{L}(f) = \frac{1}{N} \sum_{n=1}^{N} \ell(f, z_n).$$